1) a)

Let A, B and C be $(1,0,1)$, $(2,1,0)$ and $(1,2,2)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}2\\1\\0\end{pmatrix} - \begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}1\\1\\-1\end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}1\\2\\2\end{pmatrix} - \begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}0\\2\\1\end{pmatrix}$$

$$\therefore \mathbf{r} = \begin{pmatrix}1\\0\\1\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\-1\end{pmatrix} + \mu\begin{pmatrix}0\\2\\1\end{pmatrix}$$

1) b)

Let A, B and C be $(-1,2,-3)$, $(3,2,3)$ and $(-1,-1,2)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}3\\2\\3\end{pmatrix} - \begin{pmatrix}-1\\2\\-3\end{pmatrix} = \begin{pmatrix}4\\0\\6\end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}-1\\-1\\2\end{pmatrix} - \begin{pmatrix}-1\\2\\-3\end{pmatrix} = \begin{pmatrix}0\\-3\\5\end{pmatrix}$$

$$\therefore \mathbf{r} = \begin{pmatrix}-1\\2\\-3\end{pmatrix} + \lambda\begin{pmatrix}4\\0\\6\end{pmatrix} + \mu\begin{pmatrix}0\\-3\\5\end{pmatrix}$$

1) c)

Let A, B and C be $(0,0,0)$, $(3,4,1)$ and $(-9,-3,7)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}3\\4\\1\end{pmatrix} - \begin{pmatrix}0\\0\\0\end{pmatrix} = \begin{pmatrix}3\\4\\1\end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}-9\\-3\\7\end{pmatrix} - \begin{pmatrix}0\\0\\0\end{pmatrix} = \begin{pmatrix}-9\\-3\\7\end{pmatrix}$$

$$\therefore \mathbf{r} = \begin{pmatrix}0\\0\\0\end{pmatrix} + \lambda\begin{pmatrix}3\\4\\1\end{pmatrix} + \mu\begin{pmatrix}-9\\-3\\7\end{pmatrix} \Rightarrow \mathbf{r} = \lambda\begin{pmatrix}3\\4\\1\end{pmatrix} + \mu\begin{pmatrix}-9\\-3\\7\end{pmatrix}$$

2) a)

$$\begin{pmatrix} 9 \\ 18 \\ -7 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

$9 = 3 - \lambda + 2\mu \Rightarrow -\lambda + 2\mu = 6$ (1)

$18 = 2 + 2\lambda + 3\mu \Rightarrow 2\lambda + 3\mu = 16$ (2)

$\lambda = 2$ and $\mu = 4$

Check:

$-5 + 2(1) + 4(-1) = -7$

So $(9, 18, -7)$ lies on the plane.

2) b)

$$\begin{pmatrix} 24 \\ 23 \\ -17 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

$24 = 3 - \lambda + 2\mu \Rightarrow -\lambda + 2\mu = 21$ (1)

$23 = 2 + 2\lambda + 3\mu \Rightarrow 2\lambda + 3\mu = 21$ (2)

$\lambda = -3$ and $\mu = 9$

Check:

$-5 - 3(1) + 9(-1) = -17$

So $(24, 23, -17)$ lies on the plane.

2) c)

$$\begin{pmatrix} 19 \\ 5 \\ -22 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

$19 = 3 - \lambda + 2\mu \Rightarrow -\lambda + 2\mu = 16$ (1)

$5 = 2 + 2\lambda + 3\mu \Rightarrow 2\lambda + 3\mu = 3$ (2)

$\lambda = -6$ and $\mu = 5$

Check:

$-5 - 6(1) + 5(-1) = -16 \neq -22$

So $(19, 5, -22)$ does not lie on the plane.

3) a)

$$\overrightarrow{OC} = \lambda\overrightarrow{OA} + \mu\overrightarrow{OB} \Rightarrow \begin{pmatrix} 34 \\ -23 \\ 13 \end{pmatrix} = \lambda\begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} + \mu\begin{pmatrix} -6 \\ 2 \\ 3 \end{pmatrix}$$

$34 = 2\lambda - 6\mu$ (1)

$-23 = -3\lambda + 2\mu$ (2)

$\therefore \lambda = 5$ and $\mu = -4$

Check:

$5(5) - 4(3) = 13$

So $\overrightarrow{OC} = 5\overrightarrow{OA} - 4\overrightarrow{OB}$

3) b)

This tells us that O, A, B and C are coplanar.

4) a)

$$\begin{pmatrix} 3 \\ -9 \\ 8 \end{pmatrix} \cdot \begin{pmatrix} 35 \\ 33 \\ 24 \end{pmatrix} = 3 \times 35 + (-9) \times 33 + 8 \times 24 = 105 - 297 + 192 = 0$$

$$\begin{pmatrix} 3 \\ 7 \\ -14 \end{pmatrix} \cdot \begin{pmatrix} 35 \\ 33 \\ 24 \end{pmatrix} = 3 \times 35 + 7 \times 33 + (-14) \times 24 = 105 + 231 - 336 = 0$$

$\therefore 35\mathbf{i} + 33\mathbf{j} + 24\mathbf{k}$ is a normal vector to the plane.

4) b)

The plane has equation $35x + 33y + 24z = d$

$d = 35(-2) + 33(1) + 24(4) = 59$

$\therefore 35x + 33y + 24z = 59$

5)

First determine whether the point $(2, 0, 0)$ lies on the plane:

$$\begin{pmatrix}2\\0\\0\end{pmatrix} = \begin{pmatrix}0\\1\\0\end{pmatrix} + \alpha\begin{pmatrix}1\\0\\-1\end{pmatrix} + \beta\begin{pmatrix}1\\-2\\3\end{pmatrix}$$

$2 = \alpha + \beta \Rightarrow \alpha + \beta = 2$ (1)

$0 = 1 - 2\beta \Rightarrow \beta = \frac{1}{2}$ (2)

Solve (1) and (2) to get $\alpha = \frac{3}{2}$ and $\beta = \frac{1}{2}$

Check:

$0 - \left(\frac{3}{2}\right) + 3\left(\frac{1}{2}\right) = 0$ as required

So $(2, 0, 0)$ lies on the plane.

We also have that $\begin{pmatrix}-1\\2\\-3\end{pmatrix} = -\begin{pmatrix}1\\-2\\3\end{pmatrix}$ so the direction vector of the line is parallel to one of the direction vectors of the plane.

So the line lies in the plane.

6) a)

$$\mathbf{n}_1 = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\2 & 1 & 4\\3 & 1 & 0\end{vmatrix} = (1\times 0 - 4\times 1)\mathbf{i} - (2\times 0 - 4\times 3)\mathbf{j} + (2\times 1 - 1\times 3)\mathbf{k}$$

$$= -4\mathbf{i} + 12\mathbf{j} - \mathbf{k}$$

$$\mathbf{n}_2 = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & 3 & -2\\2 & 1 & -1\end{vmatrix} = (3\times(-1) - (-2)\times 1)\mathbf{i} - (1\times(-1) - (-2)\times 2)\mathbf{j} +$$

$$(1\times 1 - 3\times 2)\mathbf{k}$$

$$= -\mathbf{i} - 3\mathbf{j} - 5\mathbf{k}$$

$$\cos\theta = \frac{\begin{pmatrix}-4\\12\\-1\end{pmatrix}\cdot\begin{pmatrix}-1\\-3\\-5\end{pmatrix}}{\sqrt{(-4)^2+12^2+(-1)^2}\times\sqrt{(-1)^2+(-3)^2+(-5)^2}}$$

$$\Rightarrow \cos\theta = \frac{(-4)\times(-1)+12\times(-3)+(-1)\times(-5)}{\sqrt{161}\times\sqrt{35}}$$

$$\Rightarrow \cos\theta = -\frac{27\sqrt{115}}{805} \Rightarrow \theta = 111.080\ldots°$$

Acute Angle $= 68.9°$ (1 DP)

6) b)

$$\mathbf{n}_1 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & 0 & 2 \\ 3 & -2 & 0 \end{vmatrix} = (0 \times 0 - 2 \times (-2))\mathbf{i} - (5 \times 0 - 2 \times 3)\mathbf{j} + (5 \times (-2) - 0 \times 3)\mathbf{k}$$

$$= 4\mathbf{i} + 6\mathbf{j} - 10\mathbf{k}$$

$$\mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 1 & 1 \\ 3 & -2 & 3 \end{vmatrix} = (1 \times 3 - 1 \times (-2))\mathbf{i} - ((-2) \times 3 - 1 \times 3)\mathbf{j} + ((-2) \times (-2) - 1 \times 3)\mathbf{k}$$

$$= 5\mathbf{i} + 9\mathbf{j} + \mathbf{k}$$

$$\cos\theta = \frac{\begin{pmatrix} 4 \\ 6 \\ -10 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 9 \\ 1 \end{pmatrix}}{\sqrt{4^2+6^2+(-10)^2} \times \sqrt{5^2+9^2+1^2}}$$

$$\Rightarrow \cos\theta = \frac{4 \times 5 + 6 \times 9 + (-10) \times 1}{\sqrt{152} \times \sqrt{107}}$$

$$\Rightarrow \cos\theta = \frac{16\sqrt{4066}}{2033} \Rightarrow \theta = 59.878\ldots°$$

Acute Angle $= 59.9°$ (1 DP)

7) a)

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= (0 \times 1 - 0 \times 1)\mathbf{i} - (1 \times 1 - 0 \times 0)\mathbf{j} + (1 \times 1 - 0 \times 0)\mathbf{k}$$

$$= -\mathbf{j} + \mathbf{k}$$

The plane has equation $-y + z = d$

$d = -(1) + (0) = -1$

$-y + z = -1$

$\Rightarrow y - z = 1$

7) b)

$$\begin{pmatrix}2\\1\\1\end{pmatrix}\times\begin{pmatrix}4\\3\\1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&1&1\\4&3&1\end{vmatrix}$$

$$=(1\times1-1\times3)\mathbf{i}-(2\times1-1\times4)\mathbf{j}+(2\times3-1\times4)\mathbf{k}$$

$$=-2\mathbf{i}+2\mathbf{j}+2\mathbf{k}$$

The plane has equation $-2x+2y+2z=d$

$d=-2(1)+2(2)+2(5)=12$

$-2x+2y+2z=12$

$\Rightarrow -x+y+z=6$

7) c)

$$\begin{pmatrix}3\\2\\0\end{pmatrix}\times\begin{pmatrix}0\\-5\\6\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&2&0\\0&-5&6\end{vmatrix}$$

$$=\big(2\times6-0\times(-5)\big)\mathbf{i}-(3\times6-0\times0)\mathbf{j}+(3\times(-5)-2\times0)\mathbf{k}$$

$$=12\mathbf{i}-18\mathbf{j}-15\mathbf{k}$$

The plane has equation $12x-18y-15z=d$

$d=12(4)-18(2)-15(-5)=87$

$12x-18y-15z=87$

$\Rightarrow 4x-6y-5z=29$

7) d)

$$\begin{pmatrix}-2\\4\\-7\end{pmatrix}\times\begin{pmatrix}5\\0\\-4\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-2&4&-7\\5&0&-4\end{vmatrix}$$

$$=(4\times(-4)-(-7)\times0)\mathbf{i}-\big((-2)\times(-4)-(-7)\times5\big)\mathbf{j}+\big((-2)\times0-4\times5\big)\mathbf{k}$$

$$=-16\mathbf{i}-43\mathbf{j}-20\mathbf{k}$$

The plane has equation $-16x-43y-20z=d$

$d=-16(-2)-43(-8)-20(10)=176$

$-16x-43y-20z=176$

8) a)

$$\begin{pmatrix}2\\1\\5\end{pmatrix}\times\begin{pmatrix}-3\\-2\\9\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&1&5\\-3&-2&9\end{vmatrix}$$

$$=\big(1\times 9-5\times(-2)\big)\mathbf{i}-\big(2\times 9-5\times(-3)\big)\mathbf{j}+\big(2\times(-2)-1\times(-3)\big)\mathbf{k}$$

$$=19\mathbf{i}-33\mathbf{j}-\mathbf{k}$$

The plane has equation $19x-33y-z=d$

$d=19(1)-33(-10)-(-2)=351$

$19x-33y-z=351$

8) b)

$$\begin{pmatrix}2\\3\\-15\end{pmatrix}\times\begin{pmatrix}8\\0\\7\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&3&-15\\8&0&7\end{vmatrix}$$

$$=(3\times 7-(-15)\times 0)\mathbf{i}-(2\times 7-(-15)\times 8)\mathbf{j}+(2\times 0-3\times 8)\mathbf{k}$$

$$=21\mathbf{i}-134\mathbf{j}-24\mathbf{k}$$

The plane has equation $21x-134y-24z=d$

$d=21(0)-134(0)-24(0)=0$

$21x-134y-24z=0$

9) a)

$$\begin{pmatrix}0\\1\\3\end{pmatrix}\times\begin{pmatrix}-2\\3\\2\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\0&1&3\\-2&3&2\end{vmatrix}$$

$$=(1\times 2-3\times 3)\mathbf{i}-\big(0\times 2-3\times(-2)\big)\mathbf{j}+\big(0\times 3-1\times(-2)\big)\mathbf{k}$$

$$=-7\mathbf{i}-6\mathbf{j}+2\mathbf{k}$$

The plane has equation $-7x-6y+2z=d$

$d=-7(2)-6(1)+2(1)=-18$

$-7x-6y+2z=-18$

$$\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}-7\\-6\\2\end{pmatrix}=-18$$

$$\Rightarrow\mathbf{r}\cdot\begin{pmatrix}-7\\-6\\2\end{pmatrix}=-18$$

9) b)

$$\begin{pmatrix}20\\-8\\5\end{pmatrix}\times\begin{pmatrix}15\\8\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\20&-8&5\\15&8&0\end{vmatrix}$$

$$=\big((-8)\times 0-5\times 8\big)\mathbf{i}-(20\times 0-5\times 15)\mathbf{j}+(20\times 8-(-8)\times 15)\mathbf{k}$$

$$=-40\mathbf{i}+75\mathbf{j}+280\mathbf{k}$$

The plane has equation $-40x+75y+280z=d$

$d=-40(0)+75(-8)+280(16)=3880$

$-40x+75y+280z=3880$

$\Rightarrow -8x+15y+56z=776$

$$\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}-8\\15\\56\end{pmatrix}=776$$

$$\Rightarrow\mathbf{r}\cdot\begin{pmatrix}-8\\15\\56\end{pmatrix}=776$$

10) a)

$$\cos\theta=\frac{\begin{pmatrix}2\\-3\\8\end{pmatrix}\cdot\begin{pmatrix}-4\\1\\2\end{pmatrix}}{\sqrt{2^2+(-3)^2+8^2}\times\sqrt{(-4)^2+1^2+2^2}}$$

$$\Rightarrow\cos\theta=\frac{2\times(-4)+(-3)\times1+8\times2}{\sqrt{77}\times\sqrt{21}}$$

$$\Rightarrow\cos\theta=\frac{5}{\sqrt{1617}}\Rightarrow\theta=82.857\ldots°$$

Acute angle is 82.9° (1 DP)

10) b)

First write Π_3 in cartesian form:

$$\begin{pmatrix}2\\6\\-3\end{pmatrix}\times\begin{pmatrix}5\\4\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&6&-3\\5&4&-1\end{vmatrix}$$

$$=(6\times(-1)-(-3)\times4)\mathbf{i}-(2\times(-1)-(-3)\times5)\mathbf{j}+(2\times4-6\times5)\mathbf{k}$$

$$=6\mathbf{i}-13\mathbf{j}-22\mathbf{k}$$

The plane has equation $6x-13y-22z=d$

$d=6(3)-13(-5)-22(6)=-49$

$6x-13y-22z=-49$

Now find the angle:

$$\cos\theta=\frac{\begin{pmatrix}2\\-3\\8\end{pmatrix}\cdot\begin{pmatrix}6\\-13\\-22\end{pmatrix}}{\sqrt{2^2+(-3)^2+8^2}\times\sqrt{6^2+(-13)^2+(-22)^2}}$$

$$\Rightarrow\cos\theta=\frac{2\times6+(-3)\times(-13)+8\times(-22)}{\sqrt{77}\times\sqrt{689}}$$

$$\Rightarrow\cos\theta=-\frac{125}{\sqrt{53053}}\Rightarrow\theta=122.867\ldots°$$

Acute angle is $180°-122.867\ldots°=57.1°$ (1 DP)

11)

$$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\begin{pmatrix}-3\\1\\0\end{pmatrix}-\begin{pmatrix}0\\2\\6\end{pmatrix}=\begin{pmatrix}-3\\-1\\-6\end{pmatrix}$$

$$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=\begin{pmatrix}3\\0\\-4\end{pmatrix}-\begin{pmatrix}0\\2\\6\end{pmatrix}=\begin{pmatrix}3\\-2\\-10\end{pmatrix}$$

$$\overrightarrow{AB}\times\overrightarrow{AC}=\begin{pmatrix}-3\\-1\\-6\end{pmatrix}\times\begin{pmatrix}3\\-2\\-10\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-3&-1&-6\\3&-2&-10\end{vmatrix}$$

$$=\big((-1)\times(-10)-(-6)\times(-2)\big)\mathbf{i}-\big((-3)\times(-10)-(-6)\times3\big)\mathbf{j}+\big((-3)\times(-2)-(-1)\times3\big)\mathbf{k}$$

$$=-2\mathbf{i}-48\mathbf{j}+9\mathbf{k}$$

The plane has equation $-2x-48y+9z+d=0$

Substitute in the point $A(0,2,6)$ to find d:

$-2(0)-48(2)+9(6)+d=0\Rightarrow d=42$

So the plane has equation $-2x-48y+9z+42=0$ or $2x+48y-9z-42=0$

12)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 3 \\ 12 \end{pmatrix} - \begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \\ 11 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 6 \\ -2 \\ 10 \end{pmatrix} - \begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 9 \end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} -2 \\ 8 \\ 11 \end{pmatrix} \times \begin{pmatrix} 2 \\ 3 \\ 9 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 8 & 11 \\ 2 & 3 & 9 \end{vmatrix}$$

$$= (8 \times 9 - 11 \times 3)\mathbf{i} - \big((-2) \times 9 - 11 \times 2\big)\mathbf{j} + \big((-2) \times 3 - 8 \times 2\big)\mathbf{k}$$

$$= 39\mathbf{i} + 40\mathbf{j} - 22\mathbf{k}$$

The plane has equation $\left(\mathbf{r} - \begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix} \right) \cdot \begin{pmatrix} 39 \\ 40 \\ -22 \end{pmatrix} = 0$